

Abhishek Singh

CS undergrad at KIIT with hands-on experience building and deploying ML applications using Python, Scikit-learn, and Docker.
[gmail](#) | [portfolio](#) | [linkedin](#) | [github](#) | [itch.io](#)

EDUCATION

Kalinga Institute of Industrial Technology (KIIT)

Bachelor of Technology in Computer Science and Engineering

Bhubaneswar, India

2023 – Present | CGPA: 8.44

PROJECTS

Disease Risk Predictor | *Python, Scikit-learn, Streamlit, Docker* | [Live](#) | [GitHub](#)

2026

- Built an end-to-end ML web app predicting Diabetes and Heart Disease risk from real medical datasets
- Applied SMOTE for class imbalance, engineered interaction features, and ran GridSearchCV across 405 model fits
- Compared Logistic Regression, Random Forest, Gradient Boosting, and SVM; achieved 92.44% accuracy on heart disease
- Saved trained models and scalers as .pkl files; applied consistent StandardScaler transformation at inference
- Containerized with Docker and deployed live on Render

Financial Transaction Fraud Detection System | *Python, Scikit-learn, FastAPI, Streamlit, Docker* | [GitHub](#)

2026

- Developed an end-to-end ML system to classify financial transactions as fraudulent or non-fraudulent
- Trained a Random Forest model on transaction features such as type, amount, and balance changes
- Handled class imbalance to improve fraud detection performance
- Built a REST API using FastAPI for real-time predictions; designed interactive Streamlit frontend
- Containerized with Docker for consistent deployment

Book Recommendation System | *Python, FastAPI, Pandas, Scikit-learn, SQL* | [GitHub](#)

2026

- Built a content-based recommendation engine using TF-IDF and cosine similarity on a real-world book dataset
- Performed data preprocessing and exploratory data analysis (EDA)
- Developed REST APIs using FastAPI for recommendation serving

Game Development – IIT Guwahati Game Jam (GameDeon) | *Godot, GDScript* | [itch.io](#)

2025

- Secured **7th position among 50 teams** nationwide; invited to IIT Guwahati for final showcase
- Built and shipped a complete game as part of a 2-member team under strict time constraints

TECHNICAL SKILLS

Languages: Python, C, C++, Java, R, JavaScript

Databases & Querying: SQL, MariaDB, SQLite

Data Science: EDA, Feature Engineering, Class Imbalance Handling, Model Evaluation

ML Techniques: Random Forest, Gradient Boosting, Logistic Regression, SVM, SMOTE, GridSearchCV, TF-IDF, Cosine Similarity

Data Tools: Pandas, NumPy, Scikit-learn, Jupyter Notebook, Matplotlib, Seaborn

Frameworks & Dev Tools: Streamlit, FastAPI, Git, GitHub, Docker, Godot Engine

Web: HTML, CSS, jQuery

Platforms: GitHub Pages, Linux (CLI), VS Code, Render

CERTIFICATIONS

IBM Data Science Professional Certificate

Coursera

Courses: What is Data Science, Tools for Data Science, Data Science Methodology, Python for Data Science

Google: Foundations: Data, Data, Everywhere

Coursera

Explored data-driven decision-making, data collection, analysis, and interpretation

ADDITIONAL

Competitive Programming

Regularly practice DSA on LeetCode and CodeChef using Python and C++; focus on arrays, graphs, and dynamic programming